

Q1

1.1) Students who got this wrong generally did not know what a Nash equilibrium is, or they confused Nash equilibrium with dominant strategy equilibrium. The following are all WRONG: 1) NE is where both players get the same payoff. 2) NE is where all players are getting their best possible outcomes. 3) If a player can move so that the OTHER player can get a better payoff, it is not a NE.

1.2) Question 1.2 was not well answered. Oftentimes students did not understand what a social dilemma is. Another mistake was that students referenced question 1.1 (e.g. "It's true because the above game is a prisoner's dilemma and the outcome is suboptimal"). For those who answered correctly, the invisible hand game was frequently used as a counterexample.

1.1) Studente wat dit verkeerd gekry het, het oor die algemeen nie geweet wat 'n Nash-ewewig is nie, of hulle het Nash-ewewig met dominante strategie-ewewig verwar. Die volgende is almal VERKEERD: 1) NE is waar beide spelers dieselfde betaling kry. 2) NE is waar alle spelers hul beste moontlike uitkomst kry. 3) Indien 'n speler kan beweeg sodat die ANDER speler 'n beter betaling kan kry, is dit nie 'n NE nie.

1.2) Vraag 1.2 is nie goed beantwoord nie. Studente het dikwels nie verstaan wat 'n sosiale dilemma is nie. Nog 'n fout was dat studente na vraag 1.1 verwys het (bv. "Dit is waar, want bogenoemde speletjie is 'n gevangene se dilemma en die uitkoms is suboptimaal"). Vir diegene wat reg geantwoord het, is die onsigbare handspeletjie telkens as 'n teenvoorbeeld gebruik.

Q2

- 2.1 Take home pay / *Inkomste huis toe neem* = $150 - \text{tax } (5\% \times R150 = R7.50) = R142.50$
Employment rent per day / *Indiensnemingsopbrengs per dag* = $R142.50 - R25$
(disutility / *disutiliteit*) = $R117.50$
Looking for work / *Werk soek*: $30 \times 8 = 240$ hours/uur
Employment rent per day / *Indiensnemingsopbrengs per dag* = $R142.50 - R25$
(disutility / *disutiliteit*) = $R117.50$
Employment rent / *Indiensnemingsopbrengs* = $R117.50 - 240 = R28\ 200$
- 2.2 $R142.50 - R25 - R55 = R62.50 \times 240$
= $R15\ 000$
- 2.3 Time and effort to look for work / *Tyd en inspanning om werk te soek*
Loss of firm specific assets / *Verlies onderneming spesifieke bates*
Loss of medical aid and other benefits / *Verlies mediese fonds en ander voordele*
Social stigma of unemployment / *Sosiale stigma van werkloosheid*

Question 2.1 and 2.2 were fairly well answered. Some students mistakenly included the effort level in their calculations (usually multiplying 25 by 0.75 or calculating w/e). Another common mistake was giving the hourly or daily employment rent instead of the total. A few students

added the disutility amount to the net wage instead of subtracting it. Several students rounded their answers in the middle of the question - especially for the tax calculation ($117.5 \approx 118$). Question 2.3 was poorly answered. It was evident that most students did not study the theory for this section. Students frequently gave vague answers (e.g. the economy, inflation). Others correctly identified some factors but did not fully answer the question (e.g. “UIF” instead of “a decrease in UIF”). There were also numerous wild guesses (e.g. “Ruby commits murder” and “Children”).

Q3

Most students did well on this question. The most common mistakes was to confuse what is feasible/possible (3.2) with what is most preferred (3.3). The firm WOULD prefer point B to points A and C despite it not being possible. 3.4 asks what point is most preferred among the feasible points, because it asks about what decision the firm would make – a rational firm would not try to choose an impossible point.

“A” was not accepted for 3.5 as it does not answer the question.

Die meeste studente het goed gevaar met hierdie vraag. Die mees algemene foute was om te verwar wat haalbaar/moontlik is (3.2) met wat die meeste verkies word (3.3). Die firma SAL punt B bo punte A en C verkies ten spyte daarvan dat dit nie moontlik is nie. 3.4 vra watter punt die meeste voorkeur geniet onder die haalbare punte, want dit vra oor watter besluit die firma sal neem – 'n rasonale firma sal nie 'n onmoontlike punt probeer kies nie.

“A” is nie vir 3.5 aanvaar nie aangesien dit nie die vraag beantwoord nie.

Q4

4.1

Most students were able to answer this question correctly. The MC is the first derivative of the TC function, given by $MC = 3 + Q$. / *Die meeste studente het hierdie vraag korrek beantwoord. Die MC is die eerste afgeleide van die TC-funksie, gegee deur $MC = 3 + Q$.*

4.2

Some students struggled to derive the MR function. The first step was to write down the inverse demand function and then derive the MR function (double the slope of this demand function), given by $P = 60 - \frac{1}{2}Q$. An alternative was to calculate the TR function and find its first derivative, which is the MR function. / *Sommige studente het gesukkel om die MR-funksie af te lei. Die eerste stap was om die inverse vraagkurwe te vind en dan die MR-funksie af te lei (verdubbel die helling van hierdie vraagfunksie), gegee deur $MR = 60 - \frac{1}{2}Q$. 'n Alternatief was om die TR-funksie te bereken en die eerste afgeleide hiervan te vind, wat die MR-funksie is.*

4.3

To determine the profit-maximising price and quantity, set $MR=MC$. Thus: $3 + Q = 60 - \frac{1}{2}Q$. $Q = 38$, and $P = R50,50$. / *Om die winsmaksimerende prys- en hoeveelheid te bepaal, stel $MR=MC$. Dus: $3 + Q = 60 - \frac{1}{2}Q$. $Q = 38$ en $P = R50,50$.*

4.4

The profit margin is $P - MC$, which is $R50,50 - (3 + 38) = R9,50$. / *Die winsgrens is $P - MC$, wat $R50,50 - (3 + 38) = R9,50$ is.*

4.5

You had to calculate: $TR (P \times Q) = R50,50 \times 38 = R1919$, and $TC = 100 + 3Q + 0,5Q^2 = 100 + 3(38) + 0,5(38)^2 = R936$. $TR - TC = R983$. / *Jy moes bereken: $TR (P \times Q) = R50,50 \times 38 = R1919$, en $TC = 100 + 3Q + 0,5Q^2 = 100 + 3(38) + 0,5(38)^2 = R936$. $TR - TC = R983$.*

Q5

5.1

Most students were able to answer this question correctly but remember that MR is the EXTRA revenue a firm earns from selling one additional unit. / *Die meeste studente kon hierdie vraag korrek beantwoord, maar onthou dat MR die EKSTRA inkomste is wat 'n firma verdien uit die verkoop van een bykomende eenheid.*

5.2

Most students answered “50” and explain that it is because $MR = P$ in a perfectly competitive market. / *Die meeste studente het “50” geantwoord en verduidelik dat dit is omdat $MR = P$ in 'n volmaak mededingende mark is.*

5.3

Since firms make zero economic profit in the long run, $AC = P = 50$ / *Aangesien ondernemings in die lang termyn geen ekonomiese wins maak, is $AC = P = 50$*

5.4

Most students knew profit maximisation is at $MC=MR$, but did not know the reason. If the firm produces a lower Q, then $MR > MC$, the extra revenue obtained by producing an extra unit exceeds the extra cost the firm would be able to increase profit by increasing Q. If the firm produces at a higher Q than 100, then $MC > MR$, and the firm has been selling some units at a loss. It can increase profit by reducing production. / *Die meeste studente het geweet winsmaksimering is by $MC=MR$, maar het nie die rede geweet nie. As die firma 'n laer Q produseer, en $MR > MC$, oorskry die ekstra inkomste wat verkry word deur 'n ekstra eenheid te produseer die ekstra koste wat die firma in staat sal wees om wins te verhoog deur Q te verhoog. As die firma produseer teen 'n hoër Q as 100, dan $MC > MR$, en die firma sal sommige eenhede teen 'n verlies verkoop. Dit kan wins verhoog deur produksie te verminder.*

5.5

Shifting the MC curve down while keeping the price constant (because this firm is a price taker) means that $MR=MC$ at a higher quantity than before. / *Om die MC-kromme af te skuif terwyl die prys konstant gehou word (omdat hierdie firma 'n prysnemer is), beteken dat $MR=MC$ is teen 'n hoër hoeveelheid as voorheen.*

5.6

Students had trouble to distinguish between a firm and a market. For the market, equilibrium price will drop and equilibrium quantity increases. The firm will respond by reducing its quantity supplied. / *Studente het gesukkel om tussen 'n firma en 'n mark te onderskei. Vir die*

mark sal die ewewigsprys daal en die ewewigshoeveelheid styg. Die firma sal reageer deur sy hoeveelheid verskaf te verminder.

Q5

- 5.1 Most students were able to define the concept of marginal revenue. Both mathematical and verbal explanations were accepted.
- 5.2 Most students managed to identify the value of the MR. However, some students struggled to explain that this is due to the MR equaling the market price in perfectly competitive markets. Some students argued that it had to do with the point where $MR = MC$, which is incorrect.
- 5.3 This question was poorly answered. Since the market is in long-run equilibrium, the firm's average cost equals the market price.
- 5.4 Students were able to identify that this is the point where $MR = MC$. However, only a few students correctly explained why this is the optimal point. To get full marks for this question, students had to explain what will happen to profits if the firm produces a lower Q than 100 and what will happen if the firm produces a higher Q .
An alternative explanation is to provide the mathematical proof that profit is maximized where $MR=MC$. Some students followed this approach.
- 5.5 Although students managed to identify that the firm's quantity supplied would increase, many students said that the price will also change which is incorrect since this firm is a price-taker. The price therefore remains constant. Some students mentioned changes in the firm's market power which is incorrect.
- 5.6 This question was poorly answered. A few students managed to identify that the equilibrium quantity would increase, and the equilibrium price would decrease. Students struggled to explain what will happen to the firm's quantity as they did not recall that the decrease in the equilibrium price means that the firm now faces a lower price.

Q6

- 6.1 Total surplus: $\frac{1}{2}(4800 - 30) \times 530$
= R1 264 050
You can make different calculations by calculating producer and consumer surplus separately. You will get the same answer. / *Jy kan ander berekeninge maak deur verbruikers- en produsentesurplus apart te bereken. Jy sal dieselfde antwoord kry.*
- 6.2 $DWL_{800} = \frac{1}{2}(800 - 530) \times (4\,030 - 1\,600)$
= R328 050
Some students calculated total surplus, but that was not necessary. / *Sommige studente het die totale surplus bereken, maar dit was nie nodig nie.*

Question 6.1 was generally well answered but 6.2 was only partially well answered. Students knew how to calculate the deadweight loss but very few managed to calculate the total surplus. Very few students in general received full marks for the second question and overall few students received full marks for the whole question 6.

Q7

The question was answered quite poorly.

Graph: The majority of students drew a demand and supply curve, but many omitted the new supply curve with added tax. Points of interest on the graph (like the new equilibrium, new price paid by buyers and price received by sellers) were often not indicated on the graph. Students often failed to indicate that the demand curve was steeper than the supply curve.

Explanation: Most students correctly stated that the majority of the tax burden falls upon the consumer, but students gave nondescript explanations for the graphs, not explaining the detail on the graph (e.g. where equilibrium was and the fact that D had to be steeper than S).

Few students accurately described the changes brought about by the tax and many failed to use the graph to explain why the burden falls mostly on the consumer. Most students focused on changes in consumer and producer surplus but failed to mention what happens to price paid and price received. Most students noted the creation of a deadweight loss.

Many students drew multiple graphs to show what would be the case if either supply or demand was elastic compared to inelastic.

The demand curve must be steeper than the supply curve as it is more elastic. / *Die vraagkurwe moet steiler as die aanbodkurwe wees omdat dit elasties is.*

The tax shifts the SUPPLY curve vertically up with the size of the tax. / *Die belasting skuif die AANBODkurwe vertikaal op met die belasting.*

Must show the original equilibrium as well as the after-tax equilibrium. / *Moet die aanvanklike ewewig toon sowel as die ewewig na-belasting.*

If you drew your graph correctly the BUYER will bear most of the tax and the seller a smaller part. THIS was basically what was asked. This must be correct to get a good mark. / *Indien jou skets reg geteken is, sal die KOPER die meeste van die belasting absorbeer en die verkoper 'n kleiner deel. DIT is wat die vraag gevra het en moet reg wees om 'n goeie punt te kry.*

Some students explained a tax distribution with only ONE demand and supply curve. It is impossible to do and then you are writing nonsense. / *Sommige studente het die belastingverdeling met slegs EEN vraag- en aanbodkurwe verduidelik. Dit is onmoontlik om te doen en dan skryf jy eintlik net nonsens.*

Some students wrote in detail about the size of consumer and producer surplus. This was not asked. The division of the tax burden was asked. / *Sommige student het ook uitgebrei oor die grootte van die verbruikerssurplus en produsentesurplus. Dit is nie wat gevra is nie. Die verdeling van die belastinglas is gevra.*

Q8

The answers to this question were mixed. The majority of good answers referenced product differentiation and patents, which were well explained in relation to market power. Some students presented very similar arguments e.g. product differentiation and “new products” as two different arguments, when in fact they are the same.

Some students confused market power with bargaining power and answered the question using content from unit 6. Many students also failed to understand the difference between market share and market power. Market power is about the ability to control the price. A firm that lowers its price or offers discounts / specials / loyalty programme rewards may be able to increase its quantity sold / market share, but this is not market power. Market power is about the ability to **raise** prices.

Some students were vague in their answers and did not clearly explain how a method increased market power (e.g. “Innovation will give the firm more market power”). Other issues included discussing advertising despite the question explicitly stating not to, and misunderstanding what lobbying and collusion are.

Die antwoorde op hierdie vraag was gemeng. Die meerderheid goeie antwoorde het verwys na produk differensiasie en patente, wat goed verduidelik is in verhouding tot markkrag. Sommige studente het baie soortgelyke argumente voorgehou bv. produk differensiasie en “nuwe produkte” as twee verskillende argumente, terwyl dit in werklikheid dieselfde is.

*Sommige studente het markmag met bedingingsmag verwar en die vraag beantwoord deur inhoud uit eenheid 6 te gebruik. Baie studente het ook nie die verskil tussen markaandeel en markkrag verstaan nie. Markkrag gaan oor die vermoë om die prys te beheer. 'n Firma wat sy prys verlaag of afslag / spesiale aanbiedinge / lojaliteitsprogram belonings aanbied, kan dalk sy hoeveelheid verkoop / markaandeel verhoog, maar dit is nie markkrag nie. Markkrag gaan oor die vermoë om pryse te **verhoog**.*

Sommige studente was vaag in hul antwoorde en het nie duidelik verduidelik hoe 'n metode markkrag verhoog het nie (bv. "Innovasie sal die firma meer markkrag gee"). Ander kwessies het die bespreking van advertensies ingesluit ten spyte van die vraag wat uitdruklik gesê het om nie te doen nie, en die misverstand van wat “lobbying” en samespanning is.

Q9

9.1) Equilibrium quantity in the unregulated market is determined by $MPC=P$ giving 17 000. Most students were able to arrive at the correct answer which was read-off the graph.

9.2) Pareto-efficient quantity is given by $MSC=P$ at 6250. Most students were able to arrive at the correct answer which was read-off the graph.

9.3) A significant number of students experienced difficulties in correctly answering this question. Tip: Consider the mark allocation when answering questions as it gives you some insights on the complexity of the question, and the amount of steps/work required to get to the answer. The optimal Pigouvian tax: $5000 - 3925 = 1075$

9.4) This question was fairly tough and required students to think a bit. Unfortunately, most students struggled with this question. The minimum acceptable offer from the farmer to Fizz to ensure a Pareto-efficient outcome should equal the profit $((P-MPC)*Q)$ lost by reducing the quantity produced from 17 000 to 6250.

$$5000 - 3925 = 1075$$

$$17\ 000 - 6250 = 10\ 750$$

$$0.5 \times 1075 \times 10\ 750 = R5\ 778\ 125$$

9.5) Any one of:

- Impediments to collective action when market failures impact many people.
- Missing information about the size, source, and incidence of the external cost.
- Difficulties enforcing the bargain.
- Limited funds for the one party to compensate the other

9.1) *Ewewigshoeveelheid in die ongereguleerde mark word bepaal deur $MPC=P$ wat 17 000 gee. Die meeste studente kon by die korrekte antwoord uitkom wat van die grafiek afgelees word.*

9.2) *Pareto-doeltreffende hoeveelheid word gegee deur $MSC=P$ by 6250. Die meeste studente kon by die korrekte antwoord uitkom wat van die grafiek afgelees word.*

9.3) *'n Beduidende aantal studente het probleme ondervind om hierdie vraag korrek te beantwoord. Wenk: Oorweeg die puntetoekenning wanneer jy vrae beantwoord, want dit gee jou 'n paar insigte oor die kompleksiteit van die vraag, en die hoeveelheid stappe/werk wat nodig is om by die antwoord uit te kom. Die optimale Pigouvian belasting: $5000 - 3925 = 1075$.*

9.4) *Hierdie vraag was taamlik moeilik en het vereis dat studente 'n bietjie dink. Ongelukkig het die meeste studente met hierdie vraag gesukkel. Die minimum aanvaarbare aanbod van die boer aan Fizz om 'n Pareto-doeltreffende uitkoms te verseker, behoort gelyk te wees aan die wins $((P-MPC)*Q)$ wat verlore gaan deur die hoeveelheid geproduseer van 17 000 tot 6250 te verminder.*

$$5000 - 3925 = 1075$$

$$17\ 000 - 6250 = 10\ 750$$

$$0.5 \times 1075 \times 10\ 750 = R5\ 778\ 125$$

9.5) *Enige een van:*

- *Belemmerings tot kollektiewe optrede wanneer markmislukkings baie mense raak.*
- *Ontbrekende inligting oor die grootte, bron en voorkoms van die eksterne koste.*
- *Probleme om die ooreenkoms af te dwing.*
- *Beperkte fondse vir die een party om die ander te vergoed*